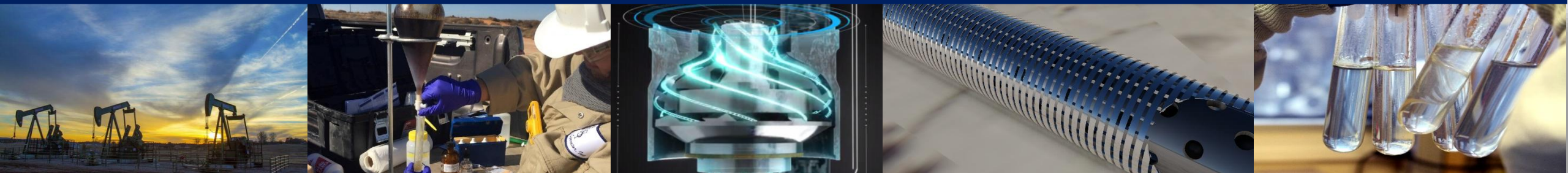
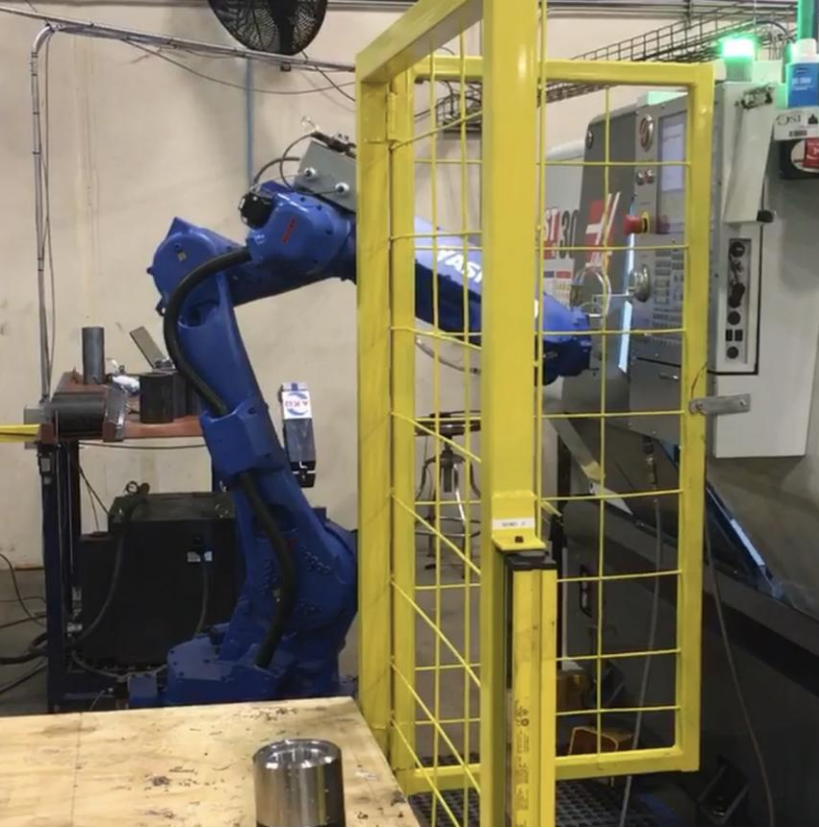




Your Source For Downhole Fluid Conditioning Systems





WHO WE ARE

Odessa Separator began in the early 1990's manufacturing gas separator for the Permian Basin oil fields. We have grown to become an industry leader in the areas of gas separation, Sand Control and Downhole chemical treatment.



OSI FACILITIES



**SAND
CONTROL**

**GAS
SEPARATION**

**CHEMICAL
TREATMENT**

[LINK](#)





Your Source For Downhole Fluid Conditioning Systems

VISION

OSI's vision is to become one of the market leaders in providing innovative solutions with new technology in areas such as Filtration / Sand Control, Gas Separation, and Chemical treatment downhole for the oil field

Core Values

Growth - **Improvement** - **Consistency** - **Creativity**
- **Justice** - **Respectful**

MISSION

Odessa Separator Inc. offers Filtration / Sand Control, Gas Separation and Chemical Treatment in downhole for the Oil Field. We are dedicated to focusing our efforts in satisfying our customers' needs with the safety of processes and protection of the environment in performing operations with an emphasis on quality and integrity in each department.



OSI

A **Leading Company** In Solids Control, Gas Separation & Chemical Treatment **Worldwide.**

INTL

Canada
Indonesia
Australia
Argelia
Oman
England
Saudi
Arabia
Romania
Colombia
Mexico
Argentina
Ecuador
Brazil
Kuwait

USA

Odessa(Principal
Office)
Nort Dakota
Oklahoma
New Mexico
California
Arkansas



Scan QR

+30 Years Of Experience

Significant Barriers to Entry

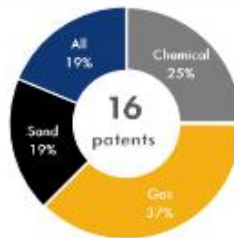


Private & Confidential

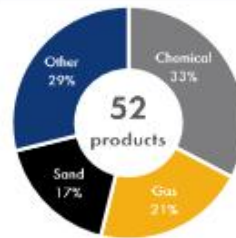
- ◆ OSI has established barriers to competition with its patented product portfolio, deep technical expertise, manufacturing and quality track record.

Patented Products & R&D

Patents Portfolio⁽¹⁾

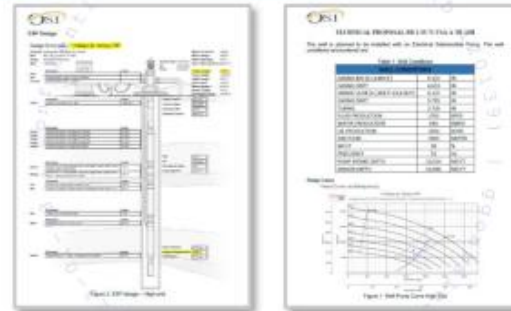


R&D Pipeline



- ◆ Protected product differentiation with 16 patents and an extensive database of drawings, design specifications and trade secrets for proprietary products.
- ◆ Dedicated R&D team comprised of experts in automation, electrical, manual machining, fabrication, patent research, design and engineering.
- ◆ Strong emphasis on identifying products that "fill a gap" in the market, which has resulted in a robust R&D pipeline of 50+ products under development or testing.

Technical Expertise



- ◆ Employs highly skilled engineers and PhD chemists who have extensive knowledge of wellbore characteristics and ancillary tools for artificial lift systems.
- ◆ Skill and knowledge of OSI's personnel enable it to design bottom hole assemblies ("BHA") that are tailored to each individual well, which improves efficiency of downhole tools and reduces the risk of failure.
- ◆ Published ~30 technical papers related to sand mitigation, gas separation and chemical treatment, among others.

Manufacturing & Quality



- ◆ Performs manufacturing at its Odessa, Texas headquarters facility containing 65,000+ square feet under roof.
- ◆ Comprehensive manufacturing capabilities include machining, plasma cutting, wire screen production, pipe perforation and welding.
- ◆ Rigorous quality assurance and control procedures have resulted in consistent and reliable product performance.
- ◆ State-of-the-art laboratory for chemical development, testing and QA/QC.



OSI TIMELINE

OSI
Foundation

New downhole chemical treatment
solutions using **micro-encapsulation**

Obtain different
patents:

1995

1996 - 2001

2012

2013

2014 -

Sand screens become the primary
focus of the company.



Gustavo Gonzalez expands
operation in Latin America



Vortex desander, two stages of filtration, dual flow
systems, and downhole chemical treatment assembly

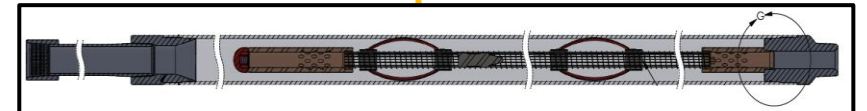
2015

2021

2022

New provisional patents on **the gas
and chemicals segments**

Obtained patents for the **ESP sand lift** &
moved to a new manufacturing facility

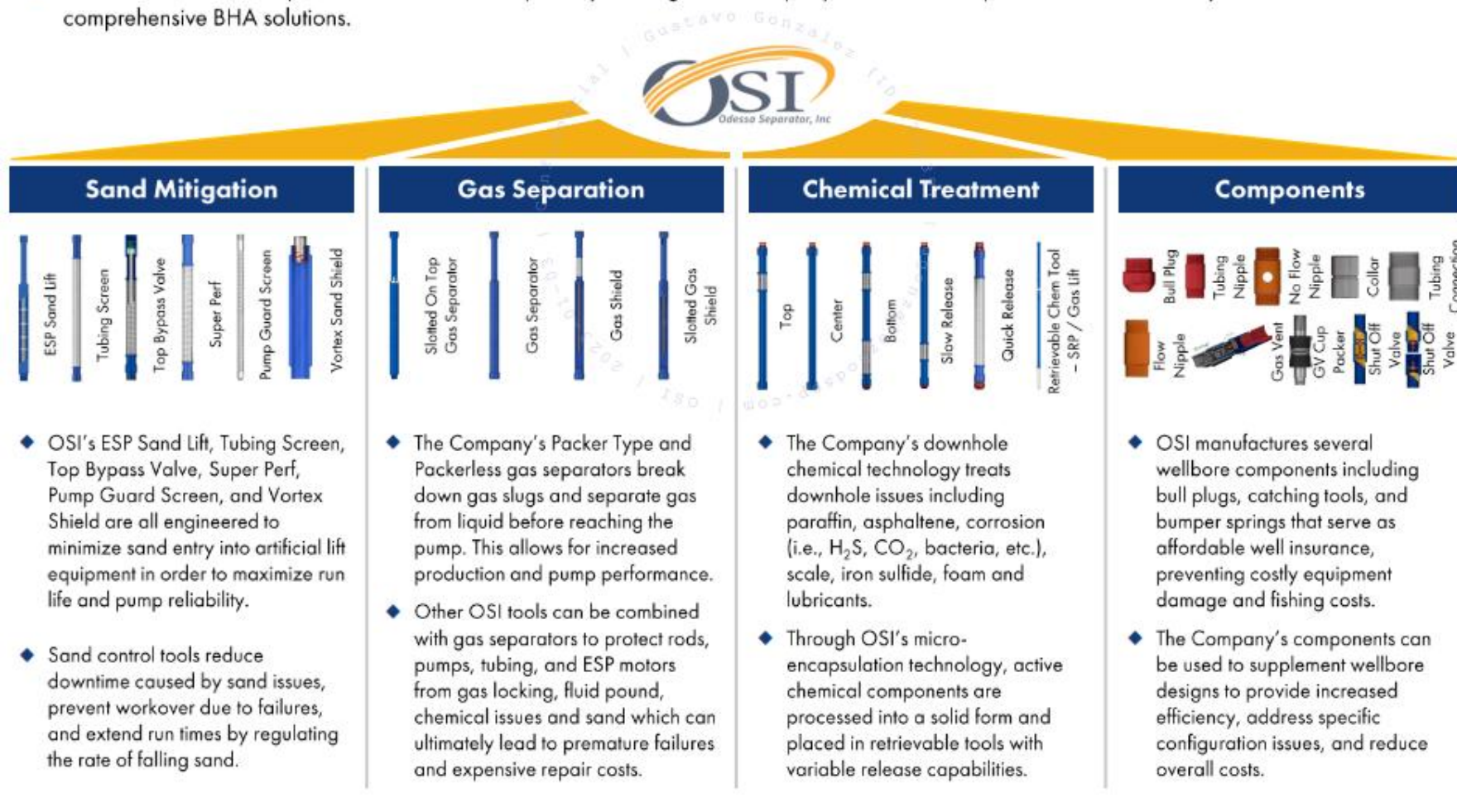


Offering Overview



Private & Confidential

- ◆ OSI maintains a broad product suite across four primary offerings. The Company sells individual products and bundled systems for more comprehensive BHA solutions.



Leading Provider of Artificial Lift Accessory Products



Private & Confidential

OSI is the only comprehensive provider of artificial lift accessory products that include sand, gas and chemical products. The depth and breadth of the Company's product offering is unmatched in the marketplace.

Providers of Artificial Lift Accessory Products							
Company	Sand Mitigation		Gas Separation		Chemical Treatment		Full Wellbore Designs
	Screen Solutions	Vortex Solutions	Packer Type Solutions	Packerless Solutions	Chemical Sticks	Slow-Release Chemicals	
	✓	✓	✓	✓	✓	✓	✓
	✗	✓	✗	✗	✗	✗	✗
	✗	✓	✓	✓	✗	✗	✓
	✓	✗	✗	✗	✗	✗	✓
	✗	✓	✓	✓	✗	✗	✗
	✗	✓	✗	✓	✗	✗	✗
	✗	✗	✗	✓	✗	✗	✗
	✗	✓	✗	✓	✗	✗	✗



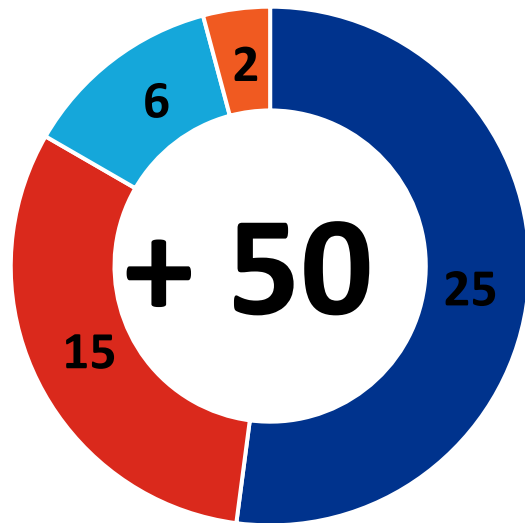
R&D Team

Patent Current Status

Application	Description	Patent
Sand Control	ESP/Rod Pump (Two Stages Filtration) Vortex/Desander	8,881,803 B1
	ESP Sand Lift "Sand Control System Above ESP"	Pending
	ESP/Rod Pump High abrasion sand control system	Pending
Chemical Treatment	Downhole chemical treatment assembly for use in a downhole wellbore	9,097,093 B1
	Method for chemically treating hydrocarbon fluid in an assembly downhole wellbore	9,097,094 B1
	Wireline Chem Tool	Pending
	Shut off Valve "Containment system for Chem applications downhole"	Pending
Gas Control	Packer type separator	Pending
	Downhole separation for well production operations Dual flow	10,415,360 B2 (OSI-4US)
	Gas Vent Tool	Pending
Others	ESP Swivel "High inclination applications"	Pending
	Downhole assembly for treating wellbore components and method for treating a wellbore	8,950,491 B2

OSI SOLUTIONS

PAPERS PUBLISHED



■ SWPSC

■ SPE

■ ALS FORUM

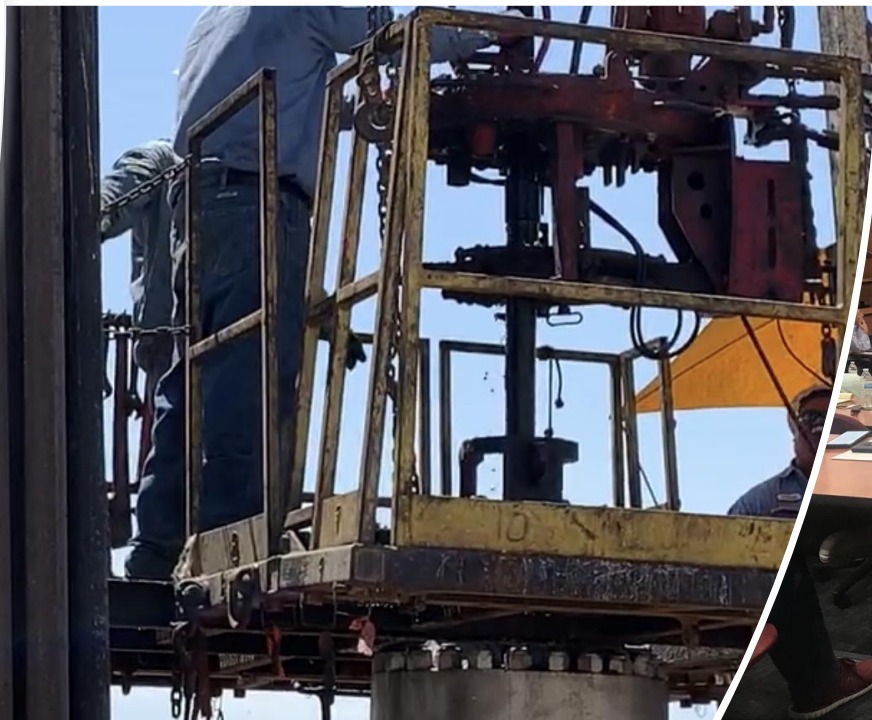
■ MEXICAN PETROLEUM CONGRESS



Homogenization Method



Successful Application of Homogenization Principle in ESPs: Case Study in South America SPE-196070-MS



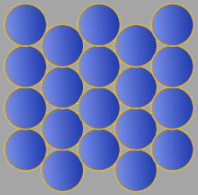
OUR VALUE PREPOSITION



OSI PRODUCTS

Below The Pump

Above The Pump



Sand Control

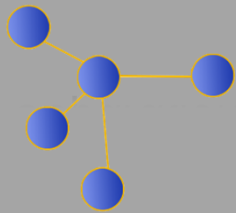
- [ESP/SRP Vortex Desander™](#)
- [ESP/SRP Screen Vortex/Desander™](#)
[\(W/Swivel, Cap String, Bypass\)](#)
- [ESP/SRP Top Bypass Valve](#)

- [ESP Sand Lift™](#)
- [ESP Swivel](#)



Gas Separation

- [ESP Vortex Regulator](#)
- [Gas Shield](#)/Slotted Gas Shield
- [ESP/SRP Guardian Shield™](#)
- [Packer Type Gas Separator](#)
- [Gas Vent™](#)



Chemical Treatment

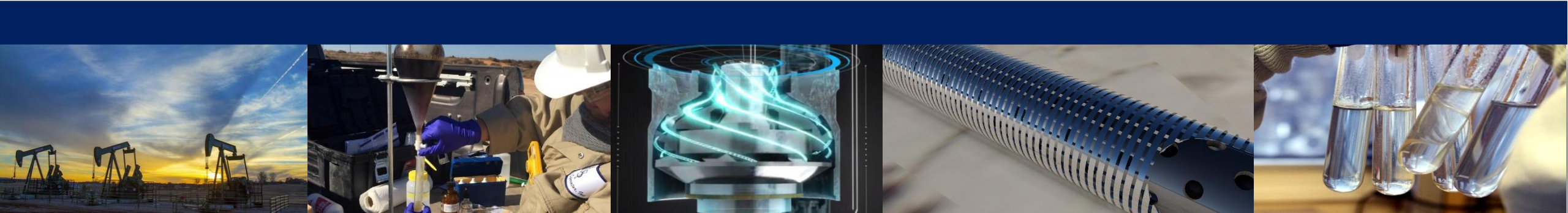
- [Chemical Screens™](#)
- [Chem Filter Tool™](#)
- [Gas Lift/SRP Retrievable Chem Tool](#)

- Chemical Screens
- Droppable Sticks



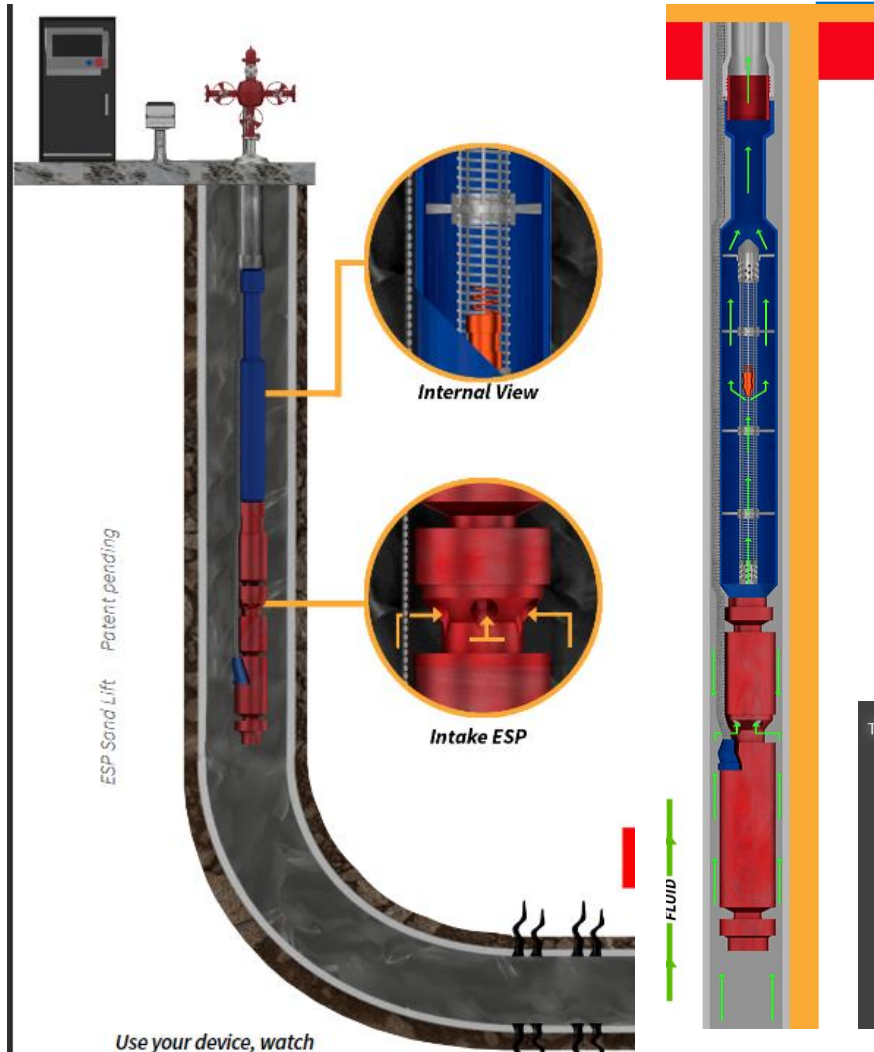
Your Source For Downhole Fluid Conditioning Systems

OSI TECHNOLOGIES

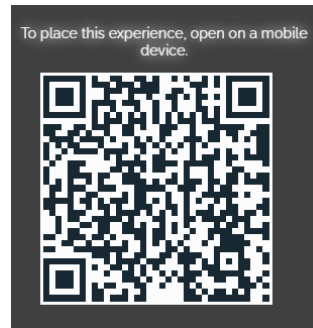


SAND CONTROL SYSTEMS

ESP SAND LIFT



Use your device, watch



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SAND CONTROL SYSTEMS

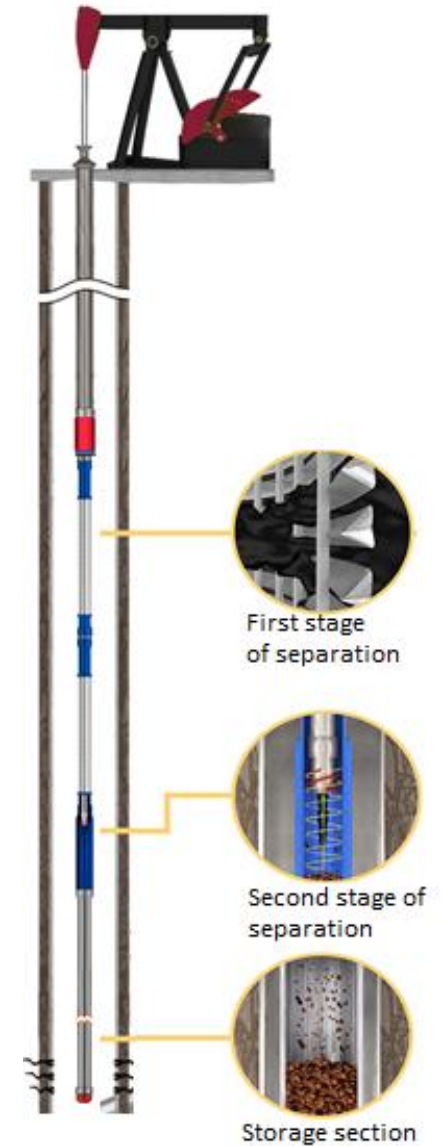
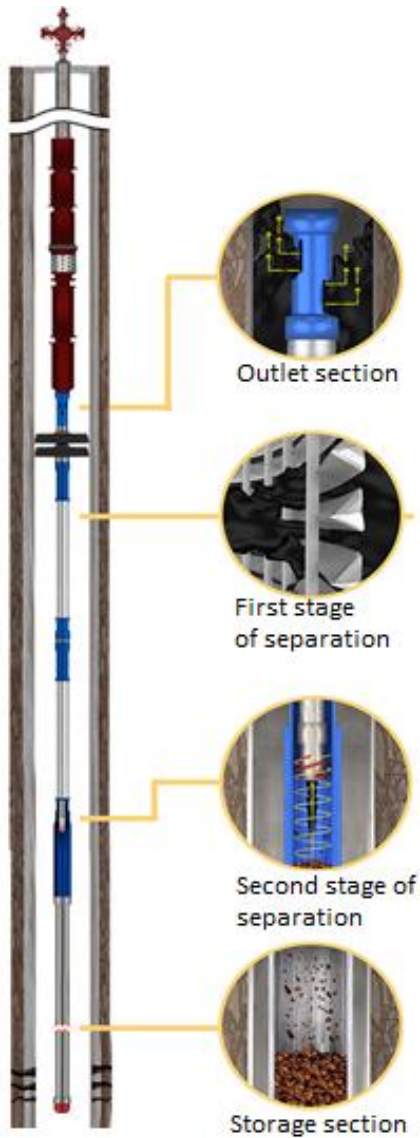
Two Separation Stages

The **first stage** utilizes either OSI's *Tubing Screen* or Super Perf. The advantage of utilizing these different tools for the intake are: filtration of larger more abrasive solids.

First stage: Range: 250 - 6730 Microns

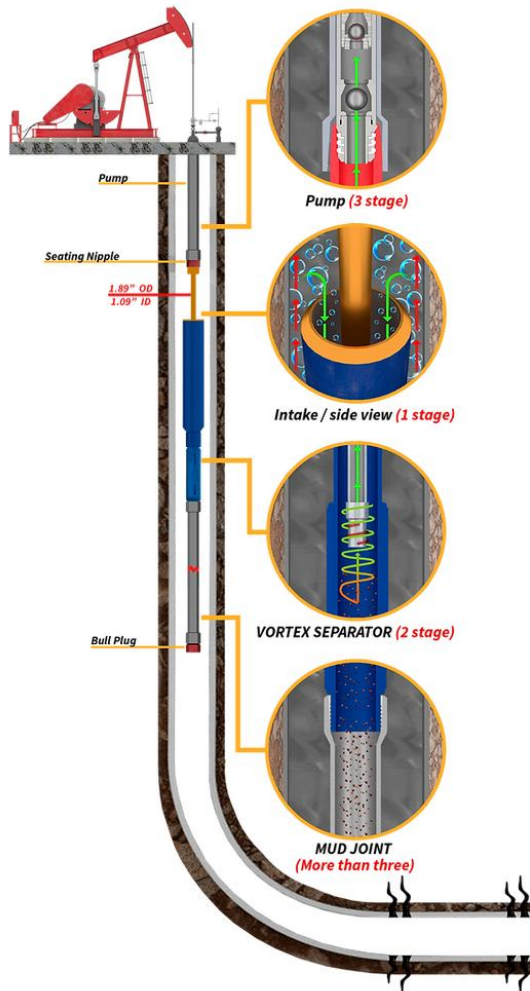
The **second stage** is the *Vortex Sand Shield* which creates a centrifugal force which separates the less abrasive solids and deposits them in the tail joint

Second stage: Range: 10 - 250 Microns



GAS SEPARATION SYSTEMS

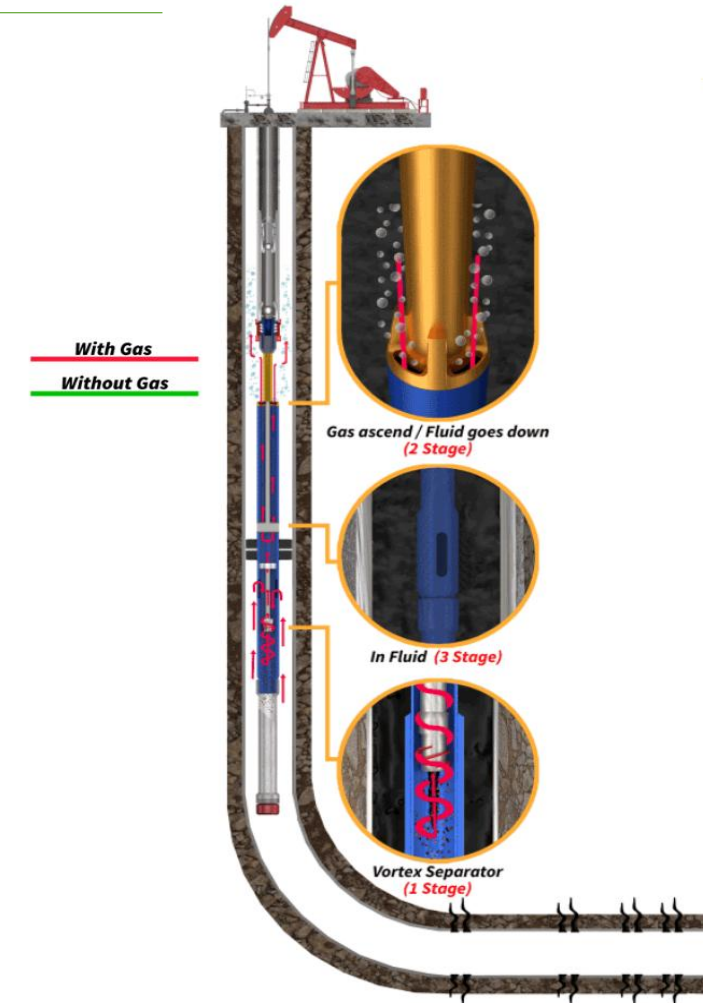
Packerless Systems



- Mitigates the gas slugs in SRP and ESP.
- Reduces or eliminates gas locking.
- Multiple stages of gas separation.
- TAC can be set below the separator.
- Utilizes both the casing and tubing as gas separators.
- Provides sand and gas separation.

- Highly efficient Gas Separator design.
- New Dual Flow System™ technology, that directs the fluid inside the tool.
- Separate the free gas to the annular section.
- Customization of the system based on the well conditions
- The outlet section is selected between slots or mesh.
- The intake section can be located to different distance from the outlet section

Packer Type Gas Separator



GAS SEPARATION SYSTEMS



"Your source for fluid conditioning systems"

SLOTTED GAS SHIELD

The Odessa Separator Slotted Gas Shield is designed specifically for wells with high lifting costs associated with gas failures. The Slotted Gas Shield is made up of diffused intake ports which minimize gas entering the separator and a large body annulus, which reduces the fluid velocity allowing for gravity driven gas separation.

The fluid enters through the slotted intake, where the first stage of separation of free gas occurs in the annular gap "by mechanical action wherein the coalescence of gas particles occurs colliding directly with the slot," then the fluid travels down inside the housing of Slotted Gas Shield.

BENEFITS

- Mitigates the gas slugs.
- Reduces or Eliminates the Gas locking.
- Multiple stages of gas separation.
- Allows sand & gas separation when is combined with the Vortex Sand Shield.

HOW IT WORKS

WITHOUT GAS
WITH GAS

Use your device by scanning the QR code



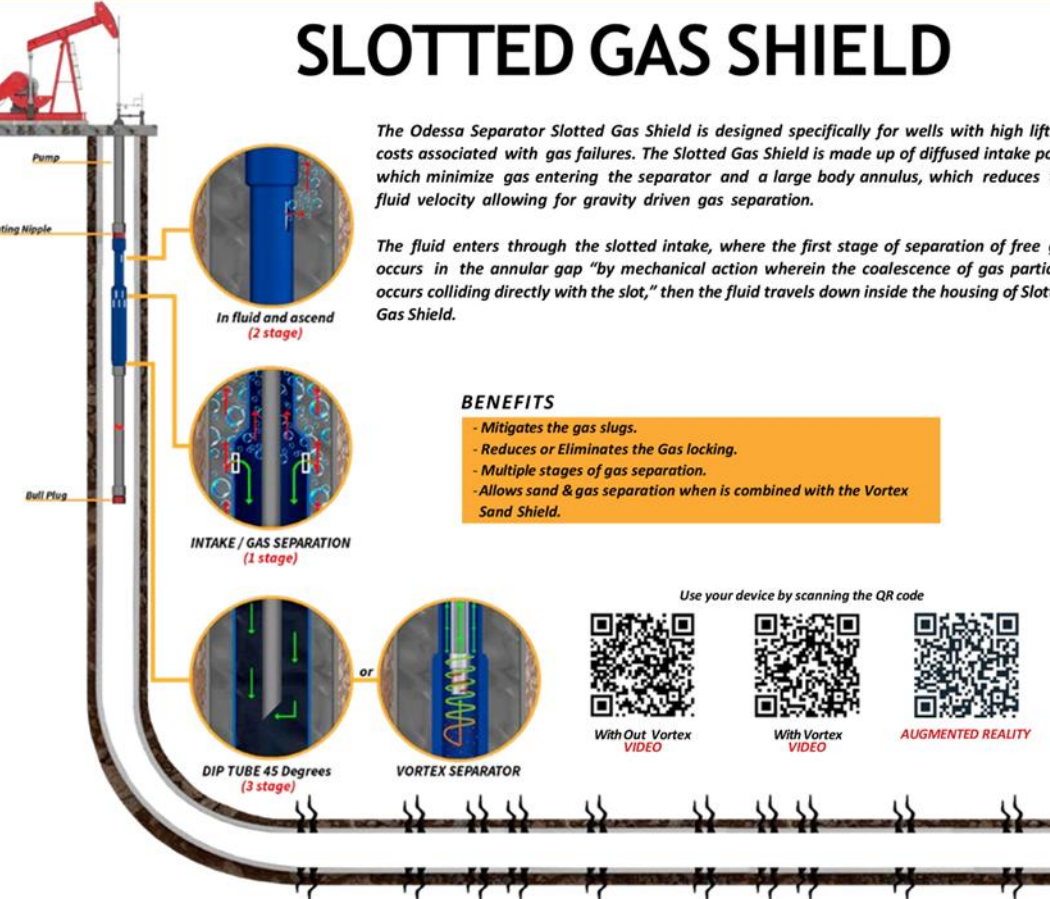
With Out Vortex
VIDEO



With Vortex
VIDEO



AUGMENTED REALITY



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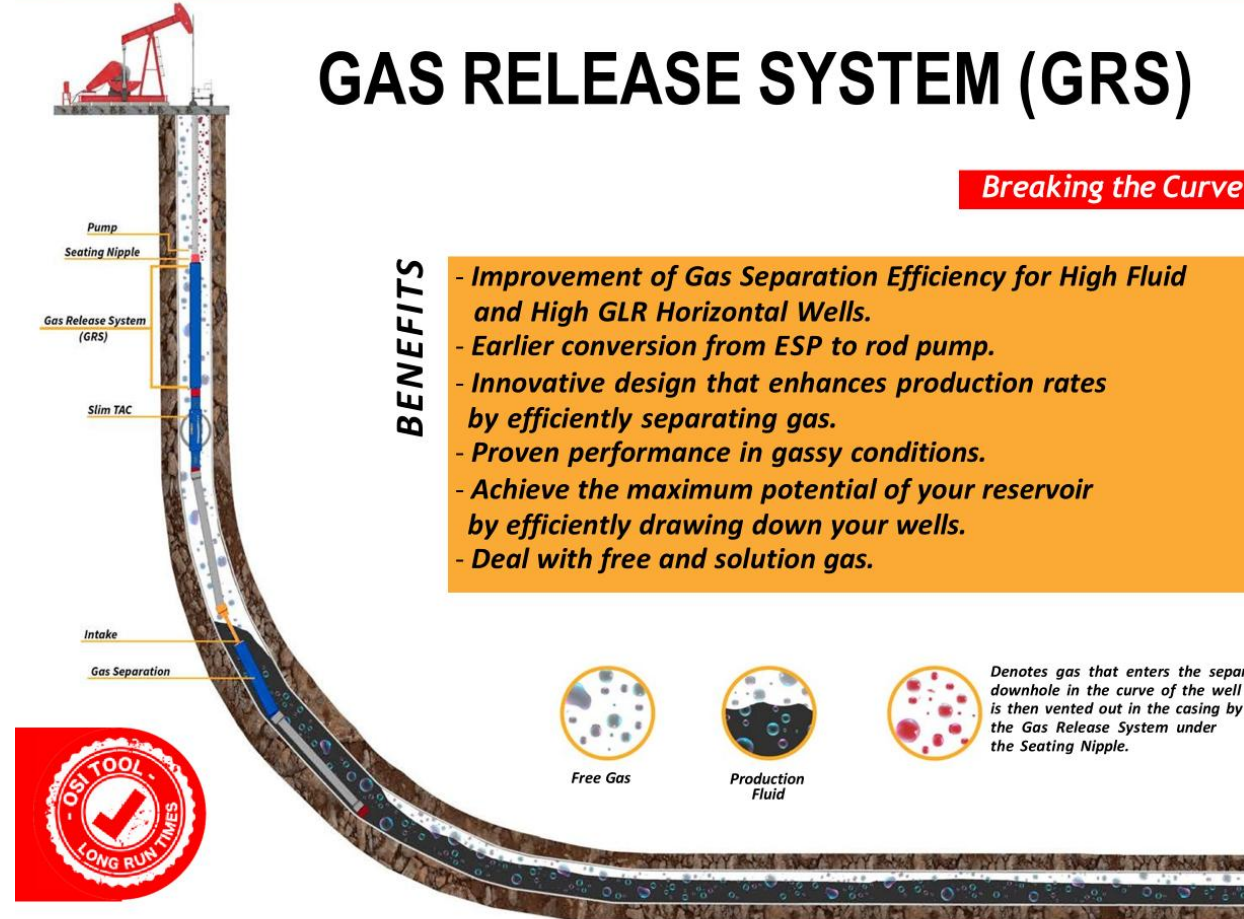
"Your source for fluid conditioning systems"

GAS RELEASE SYSTEM (GRS)

Breaking the Curve

BENEFITS

- Improvement of Gas Separation Efficiency for High Fluid and High GLR Horizontal Wells.
- Earlier conversion from ESP to rod pump.
- Innovative design that enhances production rates by efficiently separating gas.
- Proven performance in gassy conditions.
- Achieve the maximum potential of your reservoir by efficiently drawing down your wells.
- Deal with free and solution gas.



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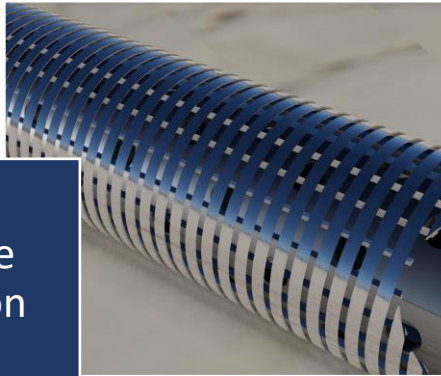
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New Gas Mitigation Technology

Design Criteria

Intake
Section



Gas.
Separator
I.D.



Tool Length



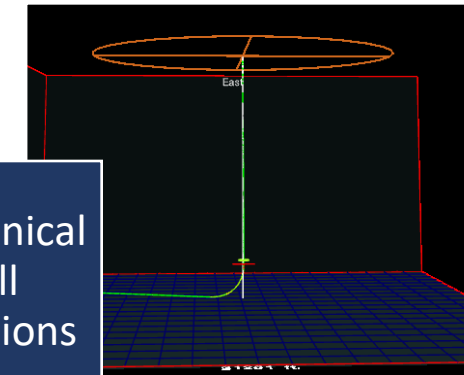
Helix Size



Shroud Size



Mechanical
well
conditions

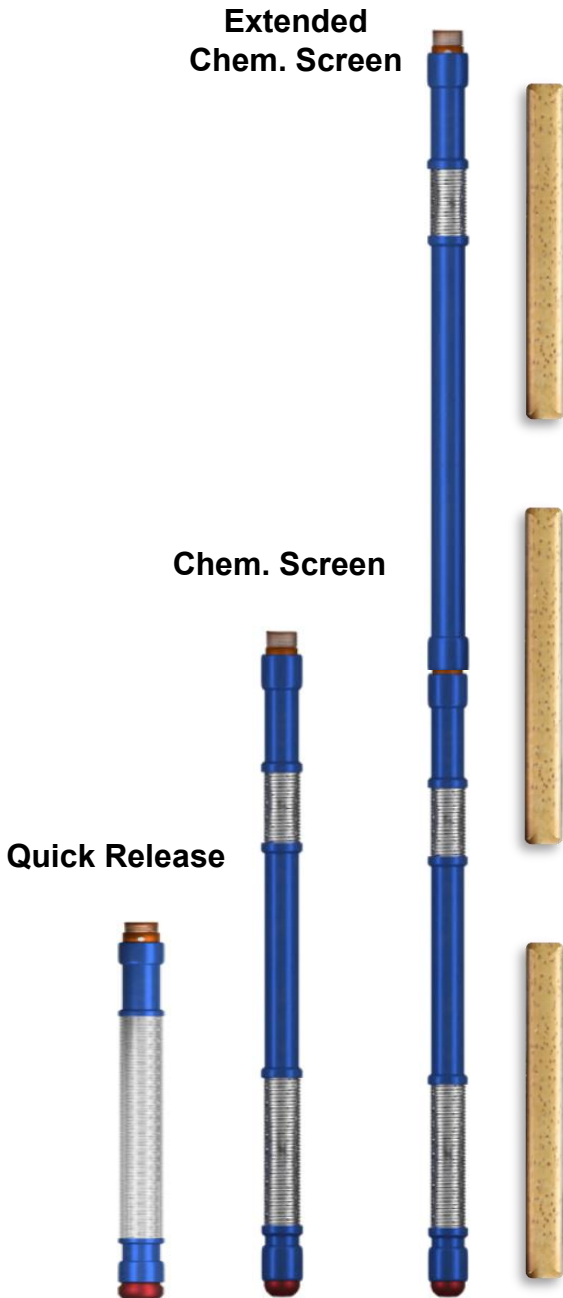


CHEMICAL TREATMENT

OSI's Inhibitors

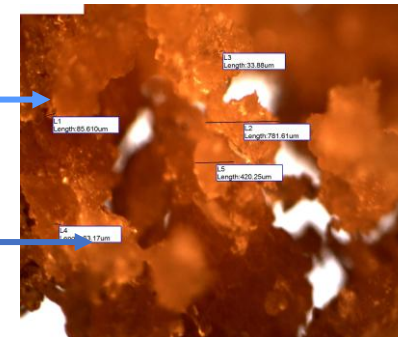
- Downhole chemical conditioning system
- Self, slow release of solid chemical with Chem Screens
- Fast release treatment using Quick Release
- Treats from bottom of wellbore, up
- Refillable tool design
- Chemical Treatment below the packer
- Compatible with any ALS
- Could be measured (Residual Test "ppm")

- Scale
- THPS/ THPC
- Scavenger
- Corrosion
- Paraffin
- Acid Surfactant
- Asphaltene
- Resins
- Defoamer
- Wax
- Biocide Applications



Encapsulated
inhibitor

Water soluble matrix
surrounding inhibitor



[Click Video](#)

* Available in 2-3/8", 2-7/8" & 3-1/2" O.D. Tools

** Longer design available for harsh conditions

MICRO-ENCAPSULATION TECHNOLOGY

Encapsulated Manufacture Process

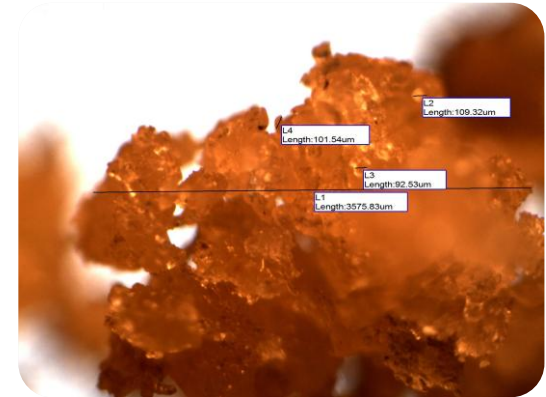


- Blended inhibitor compounds with water soluble matrix
- Extruded under pressure to form condensed chemical sticks

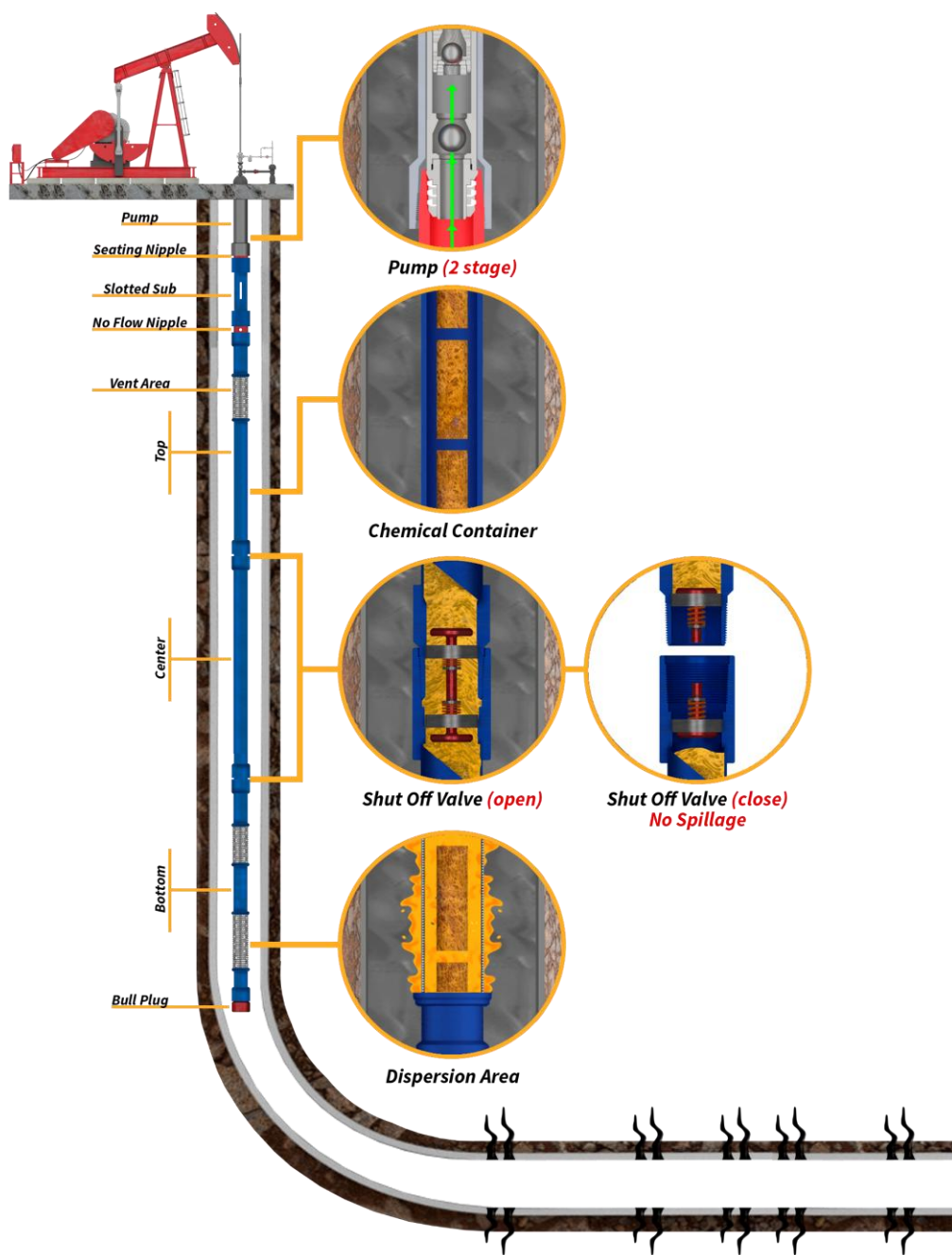


Stored and cured for
placement in Chem Tool body

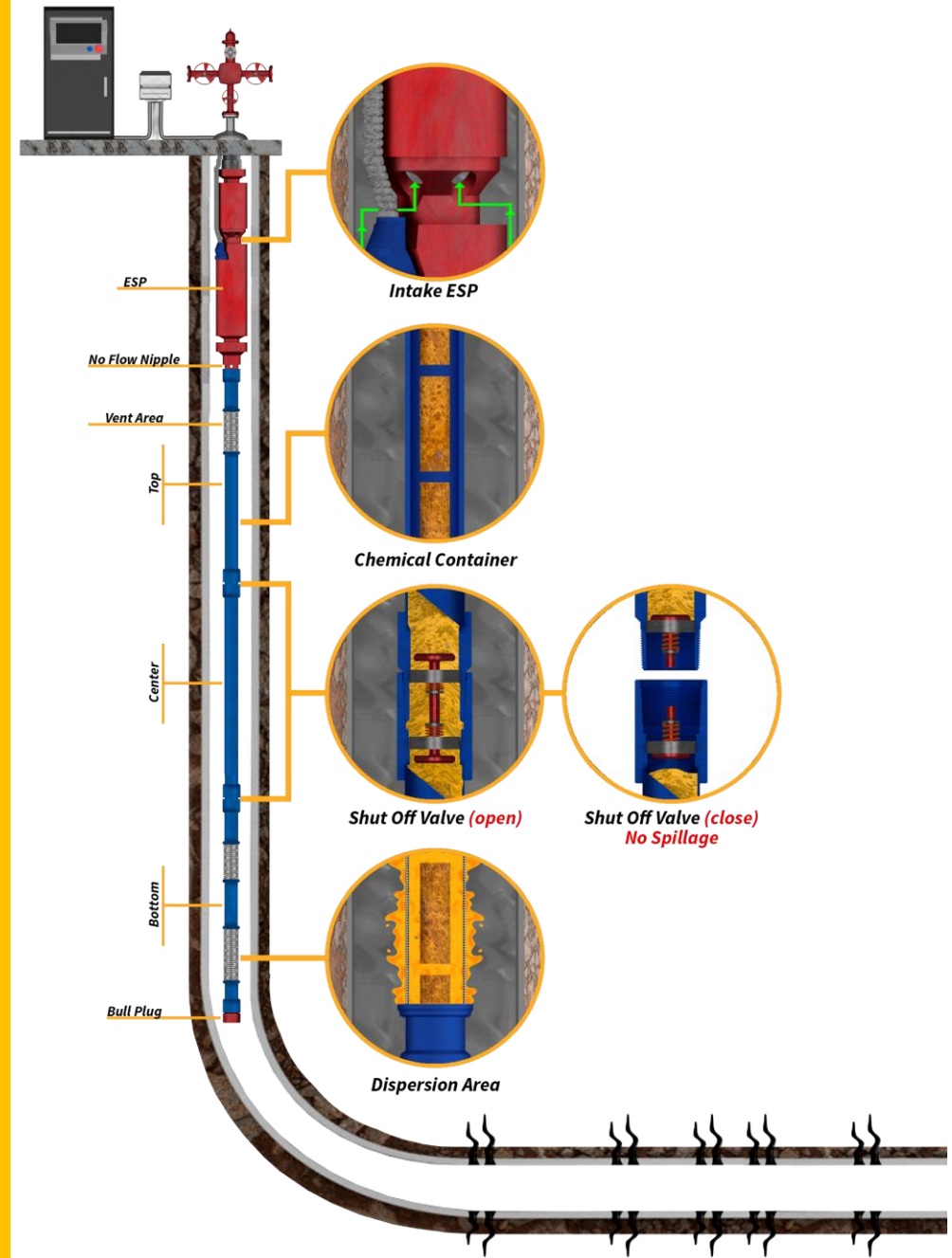
[Video](#)



SUCKER ROD PUMP



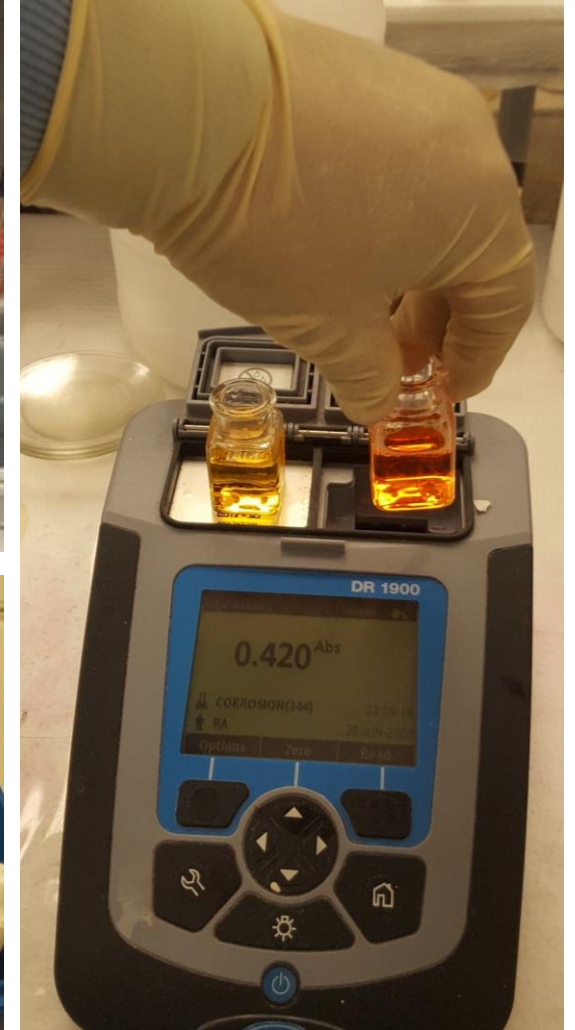
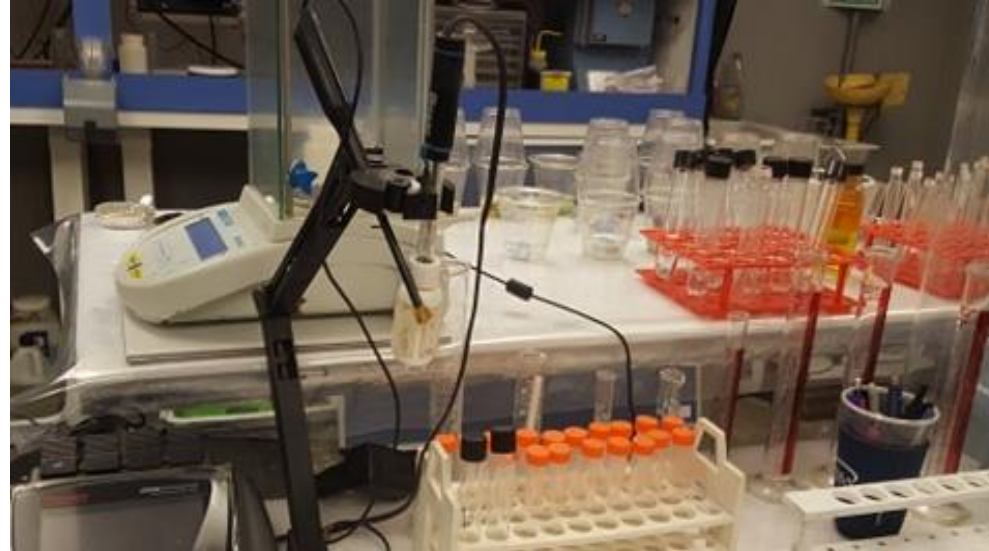
ELECTRICAL SUBMERSIBLE PUMP



TESTING SERVICES

LAB TESTING - CHEMICAL

- ✓ CWA Including
 - Iron concentration
 - Manganese concentration
 - CO₂
 - And more
- ✓ Water Residuals
 - Amines
 - Phosphates
- ✓ Polytag
- ✓ Corrosion analysis
- ✓ **Formulations Customized**



Chemical Treatment



Private & Confidential

Case Study

- ◆ OSI provides a highly effective chemical solution for producing oil & gas wells. The Company's chemical delivery system has a proven track record of performance demonstrated by the following case studies that validate the effectiveness of chemical dispersion.

Case Study: ESP Well

OSI provided chemical treatment to an ESP well that was producing nearly 1,000 barrels of fluid per day. As illustrated by the following, the Company's chemical was highly effective and exceeded minimum concentration levels for over a year. As a result of the chemical treatment, corrosion in the well was reduced by over 80%.

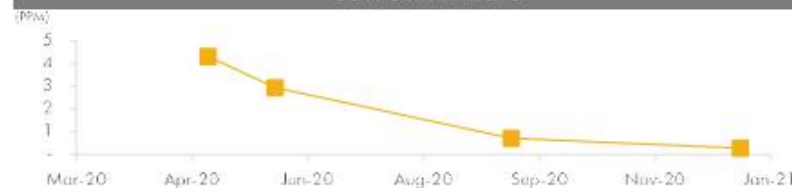
Scale & Corrosion Inhibitor



Polytag



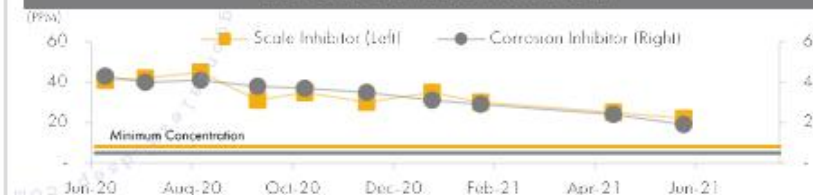
Corrosion Rate



Case Study: Rod Lift Well

OSI provided chemical treatment to a rod lift well that was producing ~150 barrels of fluid per day. As illustrated by the following, chemical levels exceeded minimum concentration levels for approximately one year, at which point the Company refilled its chemical sticks to change the mix of chemicals due to changing well conditions.

Scale & Corrosion Inhibitor



Polytag



Corrosion Rate

